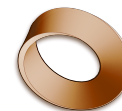


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics



1.9 – Reflection Refraction and Diffraction

Unit 1: Oscillations & Waves

Learning intentions

- a) I can define **reflection**, **refraction** and **diffraction** in terms of wave behaviour.
- b) I can explain how the change in wave speed causes refraction.
- c) I can describe how the size of a gap relative to wavelength affects diffraction.
- d) I can apply these concepts to explain why sound diffracts around corners but light does not.

Theory

When waves encounter a boundary between two different materials or an obstacle, they exhibit three main behaviours: reflection, refraction, and diffraction. **Reflection** occurs when a wave hits a barrier and bounces back into the original medium. The angle at which the wave hits the surface (angle of incidence) is equal to the angle at which it bounces off (angle of reflection). This principle allows us to hear echoes in large halls and enables technologies like sonar, where sound waves bounce off objects underwater to determine their distance.

Refraction happens when a wave passes from one medium into another with different density, causing the wave to change speed. If the wave enters the new medium at an angle (not head-on), this change in speed causes the wave to change direction. For example, when light travels from air into glass, it slows down and bends towards the normal line. Crucially, while the speed and direction of the wave change during refraction, its **frequency** remains constant because it is determined by the source of the wave.

Diffraction is the spreading out of waves as they pass through a gap or round an obstacle. The amount of spreading depends on the relationship between the wavelength of the wave and the size of the gap or obstacle. Diffraction is most pronounced when the gap width is approximately equal to the wavelength. If the gap is much wider than the wavelength, very little diffraction occurs, and the wave passes through largely unchanged.

A common real-world example involves hearing someone around a corner but not seeing them. Sound waves have wavelengths ranging from centimetres to metres, which are comparable to the size of doorways and walls, allowing sound to diffract significantly around obstacles. Visible light has a very small wavelength (around 5×10^{-7} m), so it does not diffract noticeably around everyday objects like corners or pillars, resulting in shadows rather than spread-out waves.

Complete the sentence: Diffraction is greatest when the size of the gap is approximately equal to the _____.

Word bank: frequency · wavelength · speed · reflection

Activity

Observing Wave Behaviour

You will need a ripple tank (or a tray of water with a straight wave generator), a barrier with adjustable gaps, and a ruler.

- Set up the ripple tank to produce straight waves with a known **wavelength**. Measure this wavelength in metres (m) using a ruler.
- Place a solid barrier in the path of the waves to observe **reflection**. Draw a diagram showing the incident wavefronts and the reflected wavefronts, ensuring the angles are

equal.

- Replace the solid barrier with a barrier containing a gap. Start with a gap width much larger than the wavelength and observe the pattern.
- Gradually narrow the gap until its width is approximately equal to the measured wavelength. Observe how the waves spread out into the region behind the barrier.
- Record your observations in the table below.

Gap Width (m)	Wave Behaviour Observed	Sketch of Pattern
0.1 (much larger than λ)	_____	_____
0.02 (approx equal to λ)	_____	_____

Questions

1. Define **refraction** in terms of wave speed and direction.

2. A sound wave has a frequency of 500 Hz and travels at 340 m/s. Calculate its wavelength using the equation: wavelength = speed / frequency. Give your answer in metres.

3. Explain why light waves do not diffract noticeably around a doorframe, but sound waves do.

4. When a wave undergoes refraction, which property (frequency, wavelength, or speed) remains unchanged? Justify your answer.

5. A student claims that increasing the frequency of a wave will increase the amount of diffraction through a fixed gap. Is this correct? Explain your reasoning using the relationship between frequency and wavelength.

Extension

Research how ultrasound imaging uses reflection to create images of internal body structures. Calculate the depth of a tissue layer if an ultrasound pulse (speed 1540 m/s) returns after 0.002 seconds. Remember that the time measured is for the round trip, so you must divide the total distance by two to find the depth.

